
Alien Recombination: Exploring Concept Compositions Beyond Human Cognitive Availability in Visual Art

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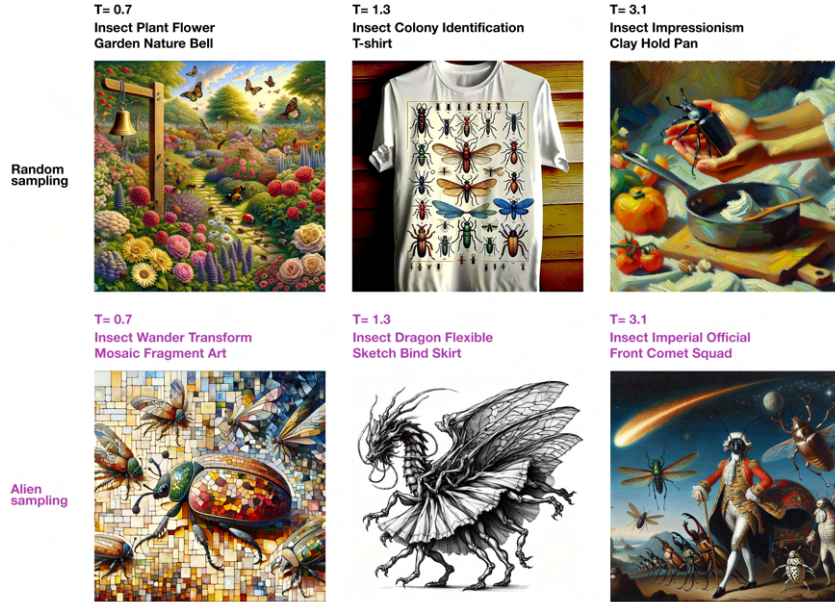


Figure 1: Comparison of images generated using random sampling versus Alien sampling with the Art model for the input sequence “Insect”. The Art model, fine-tuned on WikiArt concepts [1], combines likely concepts to produce artworks, with novelty controlled by temperature. Random sampling selects concept combinations arbitrarily, while Alien sampling ranks combinations based on cognitive unavailability and artistic fit, generating images from top-ranked sequences. For the full Alien Recombination method, see Figure 2.

1 Introduction

There is an ongoing debate about whether generative AI models can genuinely be considered creative and generate original cultural artifacts. However, the extent to which these models could unlock novel and creative connections that no human mind has conceived remains unresolved.

We hypothesize that in visual art, a vast, unexplored space of concept combinations exists, not due to inherent incompatibility, but because of the limits of the individual artist cultural horizon, including their geographical, temporal, and social embedding. For instance, the concept of “airplane” did not exist during the Renaissance period, making it cognitively unavailable to artists of that era. Today, despite familiarity with both concepts, there remains a bias against combining Renaissance style and airplanes. Cognitive science has intensively discussed human availability bias, heavily relying on immediate examples that come to mind when evaluating a specific topic [2], thereby potentially

constraining exploration of novel ideas. Although trained across temporal and cultural boundaries, generative AI models inevitably absorb human biases. As a result, these models can expect to predominantly produce cultural artifacts that align with human cognitive availability. James Evans and his team showed that counteracting such a bias could be key to algorithmic augmented scientific discovery [3, 4]. Drawing inspiration from this work, we developed a system that generates novel concept combinations in visual art by modeling and counteracting human cognitive biases. This system not only produces combinations that have never been attempted before within our dataset but also identifies and generates combinations that are cognitively unavailable to all artists in the domain.

2 Method

The Alien Recombination method employs two large language models (LLMs) to generate and rank novel combinations of artistic concepts. We extracted nine semantic-level features for each image in the WikiArt dataset [1] using CLIP [5], selecting the words most similar to the image in CLIP’s embedding space. These concepts were constrained by the WordNet Core, a curated list of the most essential words in English [6]. We then added the artwork’s style as a tenth concept from metadata.

These concepts are used to create two text datasets: the Art dataset, which contains randomly permuted concept lists for each image, and the Cognitive Availability dataset, which is created by sampling and aggregating concepts associated with individual artists to reflect their total range of ideas. Both datasets are used to fine-tune two GPT-2 models [7]: the Art model, which captures the probability of concepts co-occurring within an artwork, and the Cognitive Availability model, which assesses the probability of concepts being available within an artist’s cognitive framework.

We generate novel, cognitively unavailable combinations through a two-step process: (1) producing sequences using the Art model, and (2) ranking these sequences based on perplexity scores from both the Art and Cognitive Availability models. A lower rank in the Art model suggests a higher probability of encountering that combination in existing artworks. Similarly, a lower rank in the Cognitive Availability model indicates a higher probability of that combination being cognitively accessible. Inverting the rank of the Cognitive Availability model and fusing both ranks through a Weighted Rank Aggregation, we obtain sets of concepts that may fit great as an artwork and that are low in cognitive availability. A parameter β allows us to prioritize between these two objectives, with higher β indicating a higher emphasis on cognitive unavailability, and therefore more *alienness*. To test how the concept combinations fit in an artwork, the top-k ranked sequences are selected and visualized using a text-to-image model (DALL-E [8]), with prompts structured as `A painting that contains the concepts: <input sequence + generated sequence>`. If there is a style in the sequence, we modify the prompt accordingly.

3 Experiments and results

We compared the Alien Recombination method, using $\beta = 0.85$ for the Alien sampling, with the Art model using random sampling across temperature levels ranging from 0.1 to 3.1, with 0.3 increments. For each of the 50 inputs, 150 sequences were generated at each temperature level. From these, one sequence was selected per method, based on their respective sampling strategies.

Novelty assessment was conducted in two ways. For text-based novelty, we identified sequences with concept combinations that did not appear in any artwork within the dataset, as well as combinations that were cognitively unavailable. Image-based novelty was evaluated using GPT-4 [9] for comparative assessment, based on its alignment with human evaluators in vision-language tasks [10].

Our findings revealed that while both methods showed increased sequence novelty with higher temperatures with respect to existing artworks, the Alien sampling uniquely enhanced cognitive unavailability (Fig. 7). GPT-4 evaluations show that the Alien Recombination method consistently produced more novel images, suggesting that cognitively unavailable combinations are perceived as more innovative (Fig. 9). Additional examples and experiments are provided in the Appendix.

4 Conclusion

We present the Alien Recombination method, designed to generate concept combinations that have not previously been attempted or conceived in visual arts. Our results suggest that cognitive unavailability is a promising metric for optimizing artistic novelty, outperforming merely temperature scaling. This approach employs generative models to connect previously unconnected ideas, offering a fresh perspective on AI creativity as a combinatorial problem [11, 12]. Ethical considerations are addressed in the Appendix.

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Appendix

Architecture Overview

The Alien Recombination method is composed mainly by two GPT-2 models, the Art Model and the Cognitive Availability Model, fine-tuned in different datasets. The Art Model captures the probability of two concepts A and B being together in an artwork $P(A, B)$. The Cognitive Availability Model estimates the probability of two concepts, A and B, being cognitively available, denoted as $P(\text{cogA}, \text{cogB})$. This metric represents the probability that concepts A and B co-occur throughout an artist’s entire body of work within the dataset. Cognitive availability, in this context, is less restrictive

than co-occurrence within a single artwork because concepts A and B are considered cognitively available if they appear in different artworks by the same artist. Consequently, $P(\text{cogA}, \text{cogB})$ offers a more relaxed measure than $P(A, B)$. However, this broader scope makes it harder to identify cognitively unavailable concepts compared to identifying concepts that have never co-occurred within the same artwork. The complete workflow of the Alien Recombination method is depicted in Figure 2.

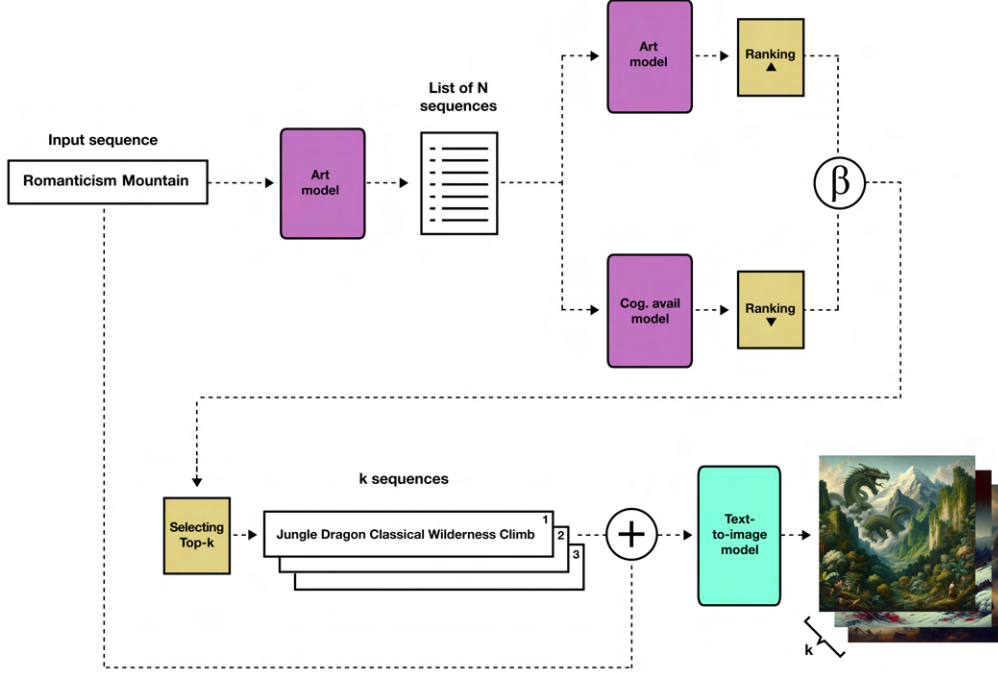


Figure 2: Schematic representation of the Alien Recombination method. The Art model generates N sequences from an input sequence. These sequences are ranked by both the Art model (ascending perplexity) and the Cognitive Availability model (descending perplexity). The ranking and selection process, termed “Alien sampling”, employs a weighted rank aggregation method parameterized by β . Increasing β prioritizes sequences that are more distant from what is cognitively available, thus enhancing *alieness*. The top- k sequences (user-defined k) resulting from this fused ranking are then processed by a text-to-image model (DALL-E [8] in this study) to generate images, using the prompt: A painting that contains the concepts: <input sequence + generated sequence>.

Experiment: Multiple images for the same input sequence

We tested the Alien Recombination method with $\beta = 0.7$ by generating various images from the same input sequence across a range of temperature values. Increasing the temperature results in more novel and diverse sequences from the Art Model providing the Alien sampling with a richer range of sequences to evaluate, allowing it to identify and return those that are more distant from what is cognitively available.

Figures 3 and 4 illustrate examples of images generated by the Alien model, using “Romanticism Mountain” and “Romanticism” as input sequences, respectively. Each example includes the sequence returned by the Alien sampling. This returned sequence, along with the input sequence, is subsequently used in the text-to-image model to produce the final images.

It is important to note that in this study the temperature parameter does not influence the model in the same way as it does in conventional language models. Using a very high temperature with the Art Model can result in sequences that include non-existent or incomplete tokens, which cannot

form meaningful artworks. To address this issue, we force the generated sequences to be composed only of words from WordNet Core, the vocabulary used to extract concepts from the artworks. This approach prevents us from introducing words that do not belong to the vocabulary of concepts we are using. In this way, we preserve the problem as pure recombinational. Therefore, we continue generating sequences until the desired number of valid sequences is obtained. This method ensures that we maintain more meaningful and stable outputs even at high temperatures, such as 3.0, unlike conventional language models used for general purposes.



Figure 3: Examples of images generated with the Alien Recombination method for the input “Romanticism Mountain”. The returned sequence by the Alien sampling is written on top of the respective image.

Discover Machinery Smoke
Industry Marsh



Keep Smoke Monk
Accumulation Monstrous



Opera Scramble Monastery
Cave Myth



Habitat Mineral Bring
Flock Reflection



Place Linger Glimpse
Midnight Fantasy



Spectrum Hemisphere Bare
Formation Laser



Fantasy Overlook Old
Quest Clue



Folk Scream Fantasy
Possession Grandfather



Notorious Imperial
Dragon Ridge



Figure 4: Examples of images generated with the Alien Recombination method for the input “Romanticism”. The returned sequence by the Alien sampling is written on top of the respective image.

Experiment: Multiple input sequences

We conducted an experiment to compare the Alien Recombination method with a baseline method for generating novel concept combinations. The baseline involved generating sequences using the Art model with increasing temperature settings to produce more random and potentially novel outputs. Specifically, we generated 50 input sequences of one or two words using the Art model with temperature 0.5 to ensure a probable initial state. We then varied the temperature from 0.1 to 3.1 in increments of 0.3 and generated 150 sequences for each temperature setting and input. For the baseline method, the candidate combination for image transformation was randomly selected, whereas the Alien Recombination method uses the Alien sampling that selects the sequence based

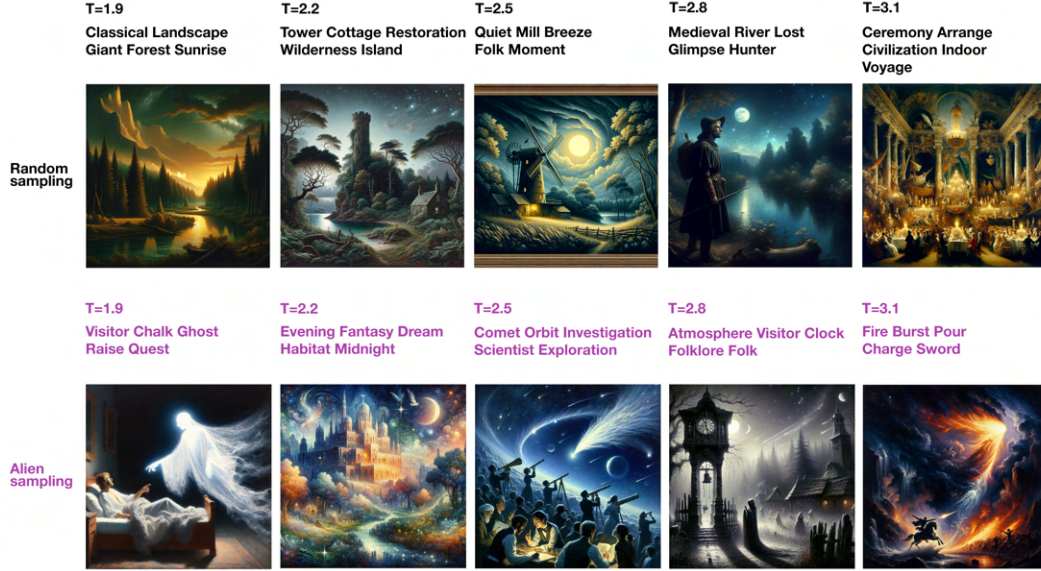
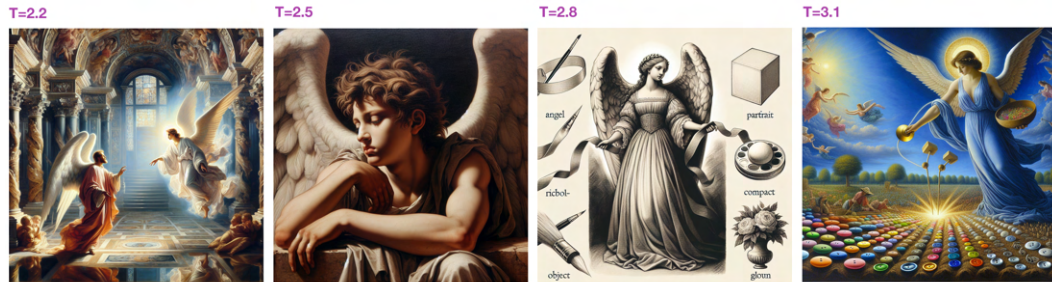


Figure 5: Comparison between images and sequences generated by the random sampling and Alien sampling for different temperature values. The input sequence is "Romanticism Night".

on its rank, computed using $\beta = 0.85$. Examples of images generated by the Alien Recombination method are shown in Figures 5 and 6, where various input sequences were used.



(a) Generated images by the Alien Recombination method given the input "Impressionism Landscape"



(b) Generated images by the Alien Recombination method given the input "High-Renaissance Angel"

Figure 6: Generated images by the Alien model for different style inputs

Text-based novelty

We evaluate the novelty of the generated combinations using two distinct metrics, both based on comparisons with the ground truth in the Art dataset:

- **Novelty relative to artworks:** It quantifies the novelty of concept combinations relative to existing artworks. Let $A = \{A_1, A_2, \dots, A_n\}$ be the set of sets, where each A_i represents the set of concepts in a single artwork within the Art dataset. Let S be the set of concepts in the generated sequence. We define the novelty measure N_{art} as:

$$N_{\text{art}} = \min(|S \setminus A_1|, |S \setminus A_2|, \dots, |S \setminus A_n|) \quad (1)$$

Where " $\setminus$ " denotes the set difference operation. This measure represents the minimum number of concepts in S that do not appear together in any single artwork, when compared individually to each artwork in the dataset. A more intuitive way to interpret this measure is that it quantifies the number of distinct concepts in the generated artwork compared to the most similar artwork in the dataset.

- **Novelty relative to cognitive availability:** It measures the novelty of concept combinations relative to individual artists' cognitive availability. Let $C = \{C_1, C_2, \dots, C_m\}$ be the set of sets, where each C_i represents the set of unique concepts used by artist i in the Art dataset. We define the novelty measure N_{cog} as:

$$N_{\text{cog}} = \min(|S \setminus C_1|, |S \setminus C_2|, \dots, |S \setminus C_m|) \quad (2)$$

This measure represents the minimum number of concepts in S that do not appear together in any single artist's cognitive framework, when compared individually to each artist's set of concepts. Thus, N_{cog} quantifies the number of distinct concepts in the generated artwork compared to the most similar artist set.

Thus, N_{art} and N_{cog} provide conservative estimates of sequence novelty, measuring the minimum number of concepts that are guaranteed to make the combination unique, regardless of which artwork or artist we compare against. These complementary measures provide a nuanced evaluation of the Alien Recombination method's ability to generate novel artistic combinations, focusing on both overall artistic innovation and the method's potential to expand individual cognitive boundaries.

For each sequence selected by both methods, we calculate N_{art} and N_{cog} . Figure 7 illustrates the average N_{art} and N_{cog} values for various temperature levels. Our findings reveal that merely increasing the temperature of the Art model suffices to generate entirely novel combinations absent from any artwork in the dataset (Figure 7a). However, while higher temperatures in the Art model can enhance novelty, they do not reliably produce combinations that are cognitively inaccessible to all artists. In contrast, the Alien Recombination method explicitly searches for cognitive unavailability in its sampling process, resulting in a higher likelihood of generating such combinations (Figure 7b).

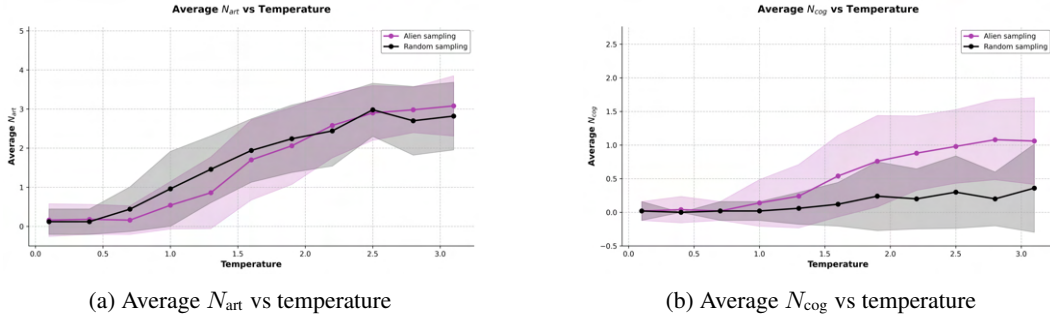
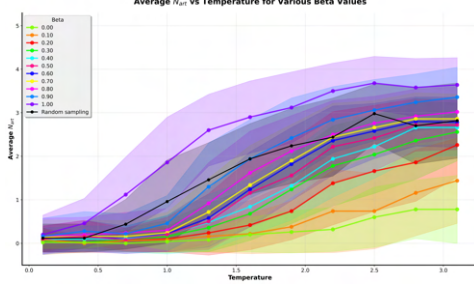


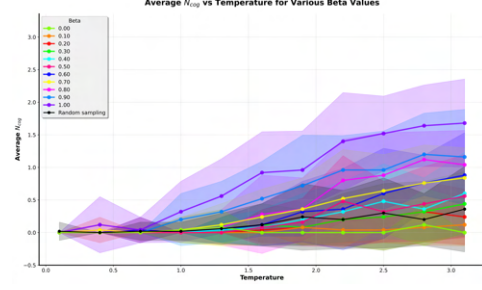
Figure 7: Comparison on the novelty of the generated sequences with respect to the artworks and the cognitive availability

Additionally, for each input and temperature level, we applied the Alien Recombination as before but gradually increasing β in increments of 0.1, from 0 (favoring the most probable concept sequence in an artwork) to 1 (favoring the most cognitively unavailable sequence).

The results in Figure 8 show that increasing β leads to more novel sequences in terms of both artistic novelty (N_{art}) and cognitive unavailability (N_{cog}). While $\beta = 1$ consistently outperforms random sampling in artistic novelty, other β values initially show lower artistic novelty due to the weight of the Art model perplexity. As temperature increases, the trade-off between cognitive unavailability and artistic fit becomes less pronounced.



(a) Average N_{art} vs temperature for multiple values of β



(b) Average N_{cog} vs temperature for multiple values of β

Figure 8: Comparison on the novelty of the generated sequences with respect to the artworks and the cognitive availability for multiple values of β

Image-based novelty

To evaluate the novelty of the images generated from the sequences selected by both sampling methods, we used GPT-4 [9]. Given that GPT-4 has been shown to align closely with human evaluators in general vision-language tasks [10], we employed it to determine which image is more novel. For each temperature and input, we compared the images generated using a random sampling, the baseline method, and the Alien sampling. We prompted GPT-4 with the instruction: As an art expert, please write a sentence indicating which image is more novel, focusing on concept combination novelty, and selected the more novel image based on GPT-4’s response. This process revealed that the images generated by the Alien sampling consistently exhibited greater novelty than those produced by the random sampling for the same input and level of temperature, as illustrated in Figure 9.

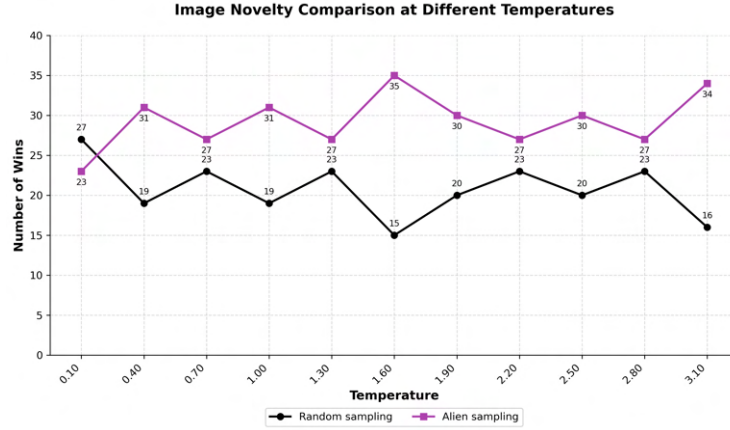


Figure 9: GPT-4 evaluation on image novelty.

On the other hand, the appeal of generated images may be influenced by the cognitive availability of the underlying conceptual combinations. We suggest that the relationship between cognitive unavailability and aesthetic appeal may follow a Wundt curve [13]. Specifically, as the cognitive unavailability of an artwork increases from low to moderate levels, its appeal tends to increase. However, excessively high levels of cognitive unavailability can lead to decreased pleasantness.

This pattern may be explained in part by the role of habituation in aesthetic responses. Habituation, a reduction in response to a stimulus after repeated exposure, can shape how appealing an artwork is perceived at first [14]. Conceptual combinations that are considered highly unusual or strange may be initially rejected by viewers due to limited prior exposure and the cognitive effort required to process such unfamiliar content.

We hypothesize that by adjusting the parameter β in the Alien sampling, it could be possible to balance between cognitively unavailable combinations and those that fit well within the context of existing artworks. We instructed GPT-4 with: As an art expert, please write a sentence indicating which image is more appealing, focusing on concept combination fitting. The results showed that the Art model with random sampling (generating relatively low cognitive unavailable combinations) outperformed the Alien sampling method with a high β (producing higher cognitive unavailability) in optimizing the overall appeal of the generated images (see Figure 10).

Nevertheless, we consider this topic requires further research. We intend to explore the impact of β on appeal and novelty more deeply, and enhance the alignment of our evaluations by incorporating assessments with human evaluators in future studies.

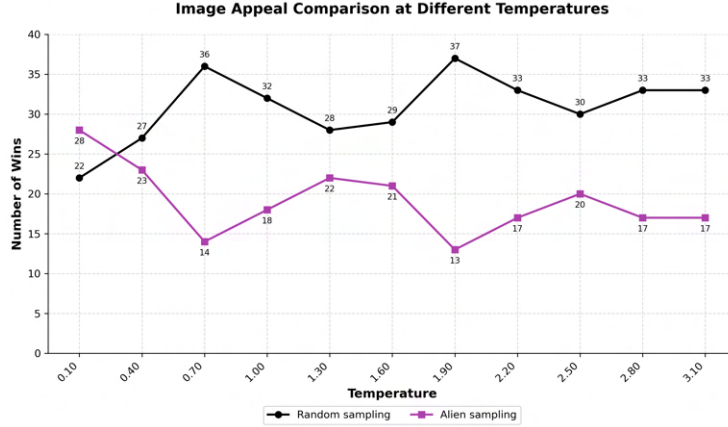


Figure 10: GPT-4 evaluation on image appeal.

Ethical Implications

This method can serve as a tool for discovering new connections in art that might be overlooked due to cultural, temporal and spatial gaps. However, the Alien Recombination method currently proposes combinations based on its training dataset, which may be biased towards 20th century Western visual art. Additionally, notions of appeal and novelty can vary depending on an individual’s background, so the model’s optimization for appeal and novelty might be skewed towards the preferences of the dataset’s demographic.

It is crucial to acknowledge that our method inherits the biases present in its constituent models, CLIP and DALL-E. During our experiment with multiple inputs, one of the generated inputs was the sequence “Indigenous Chief”. As illustrated in Figure 11, CLIP’s interpretation of “Indigenous” manifested through a diverse range of paintings, reflecting its own biases in concept recognition. Conversely, DALL-E depicted the concept “Indigenous” predominantly through a narrow, stereotypical lens of Native American imagery.

In addition to inheriting biases from the models composing the Alien Recombination method, it is essential to understand that the notion of cognitive availability is also biased. The method’s perception of what is cognitively available is heavily influenced by its training data, which is based on what is depicted in paintings and may not accurately represent reality. Consequently, the method might produce combinations that are categorized as cognitively unavailable, such as “Indigenous Computer” shown in Figure 11. Although these combinations may be underrepresented or missing in the dataset, they are entirely plausible and relevant in today’s society. This highlights the risk that concepts currently recognized in society may be misrepresented or outdated in the dataset due to biases in the training data.

WikiArt
images
with the
concept
“Indigenous”



T= 2.5

Indigenous Chief Glance Roof
Recognise Overall Rural



T= 2.8

Indigenous Chief Auction
Traditional Costume Feather Stick



T= 3.1

Indigenous Chief Stewar
Immigration Return Voluntary

Random
sampling



T= 2.5

Indigenous Chief Civic
Spokesman Lost Charter Art



T= 2.8

Indigenous Chief Mosaic
Computer Distribute Seize Native



T= 3.1

Indigenous Chief Imperial
Cable Aircraft Helicopter Warrior

Alien
sampling



Figure 11: Comparison of images that contains the concept “Indigenous” in the WikiArt dataset with the images generated using the random sampling and Alien sampling.